UNIVERSITY OF AMSTERDAM

REPORT

Analysis and Extensions of the SIR Model

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1 Introduction

Infectious diseases have had a major impact on human populations throughout history. Being able to understand how a disease spreads through a population is therefore important for predicting the development of an epidemic and controlling its damage. However, the spread of a disease depends on many different factors, such as the number of susceptible individuals, how often people come into contact with each other and how quickly infected individuals recover. Mathematical models provide a way to simplify these interactions and study how they affect the behaviour of an epidemic.

One of the most well-known models used in epidemiology is the SIR model, originally developed by Kermack and McKendrick [3]. The model divides a population into three compartments: susceptible (S), infected (I) and recovered (R). Individuals transition between these compartments as the disease spreads through the population. Although it may seem to have relatively simple structure, the SIR model can already reproduce important characteristics of an epidemic, such as the initial growth of infections, the peak number of infected individuals and the eventual decline of the disease. The model can also be extended by changing its assumptions or introducing additional compartments and parameters. This allows more complicated effects to be measured.

The purpose of this report is to investigate how the behaviour of an infectious disease changes as we vary the parameters and as the basic SIR model gets more complex. We first analyse the standard SIR model and examine the conditions under which an epidemic can occur. We then try to fit this model onto real data from an influenza outbreak in a boys' school. The model is then extended by including demography, allowing an endemic state and oscillatory behaviour to occur. Fourier analysis is used to investigate the frequency of oscillations. Finally, further variations of the SIR model are explored.

2 The SIR Model

2.1 Model Definitions and Assumptions

In the SIR model, the population is divided into three groups: susceptible (S), infected (I), and recovered (R). A susceptible person is someone who can catch the disease and move into the infected group. An infected person can then recover and move into the recovered group. In the basic SIR model, recovered people cannot become susceptible again.

For this model, we assume that the population is closed and has a constant size. This means that births, deaths and migration are not included. Therefore, when the compartments are represented as fractions of the total population,

$$S + I + R = N = 1$$

Since the total population does not change over time,

$$\frac{dN}{dt} = 0$$

We also assume that people in the population mix randomly. This means that every susceptible person has the same chance of coming into contact with an infected person. Finally, the transmission and recovery rates stay constant throughout the simulation. These assumptions make the model easier to analyse, but they also mean that it cannot represent every part of a real epidemic.

2.2 Differential equations

The movement between the three compartments can be described using a set of ordinary differential equations (ODEs). In this report, β represents the transmission rate and γ represents the recovery rate. The basic SIR model is given by

$$\frac{dS}{dt} = -\beta SI$$

$$\frac{dI}{dt} = \beta SI - \gamma I$$

$$\frac{dR}{dt} = \gamma I$$

The first equation describes the change in the susceptible population. The term βSI represents new infections from susceptible people meeting infected people. Since these people leave the susceptible group, this term is negative.

The infected group increases by the same number of new infections. At the same time, infected people recover at a rate of γI. Therefore, the infected population can either increase or decrease depending on which of these two terms is larger. The recovered group only increases, since recovered individuals do not return to the susceptible group in the basic model. If the recovery rate has the value γ, it means it takes $\frac{1}{\gamma}$ days to recover.

2.3 Basic Reproduction number

An important value in the SIR model is the basic reproduction number, R_0. It describes the expected number of new infections caused by one infected individual when almost the whole population is susceptible.

We can find the condition for an epidemic by looking at the infected equation:

$$\frac{dI}{dt} = \beta SI - \gamma I$$

For the number of infected people to increase, we need the rate of infected people to be positive.

$$\frac{dI}{dt} > 0$$

Therefore,

$$\beta SI - \gamma I > 0$$

Dividing by I gives

$$\beta S > \gamma$$

At the start of an epidemic, almost the entire population is susceptible, so $S \approx 1$. This gives

$$\frac{\beta}{\gamma} > 1$$

The basic reproduction number for this model is therefore

$$R_0 = \frac{\beta}{\gamma}$$

When $R_0 > 1$, the number of infected people initially increases and an epidemic occurs. When $R_0 < 1$, infected people recover faster than new infections are produced, so the infection dies out.

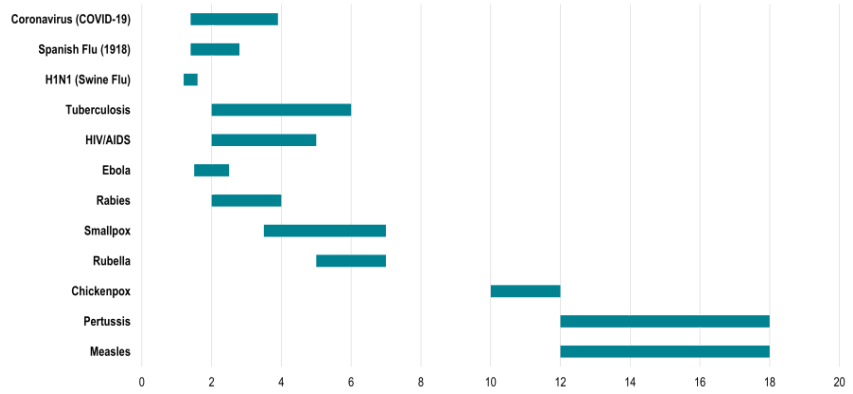


Figure 1: Reproduction Number of Infectious Diseases |The Geography of Transport Systems, 2021) [1]

The value of R_0 can differ greatly between infectious diseases. Figure 1 shows the estimated basic reproduction numbers for several well-known diseases. Diseases with a higher R_0 can generally spread more easily through a susceptible population. One of the main goals during an epidemic is to reduce the spread of disease. This can be done by reducing the effective reproduction number, for example by lowering transmission through vaccination or reducing contact between susceptible and infected people.

3 Numerical analysis of the SIR model

3.1 Numerical implementation

The differential equations from Section 2 do not have to be solved by hand. Instead, we can solve them numerically using Python. For the simulations in this report, the SIR model was written as a function which takes the current values of S, I, and R, and returns their rates of change over some period of time.

```
def sir_model(y, t, beta, gamma):
    S, I, R = y

    dSdt = -beta * S * I
    dIdt = beta * S * I - gamma * I
    dRdt = gamma * I

    return [dSdt, dIdt, dRdt]
```

Figure 2. Python function used to define the differential equations of the SIR model.

The function in Figure 2 directly follows the three differential equations introduced in Section 2. The values of S, I, and R are stored in the state of the model. The parameters β and γ control the

transmission and recovery rates. The function returns dS/dt, dI/dt, and dR/dt, which can then be numerically integrated over time.

For both simulations, the initial conditions are:

$$S(0) = 0.999, I(0) = 0.001, R(0) = 0$$

This means that initially 0.1% of the population is infected, while the remaining 99.9% is susceptible. Nobody has recovered yet. These initial conditions also satisfy:

$$S(0) + I(0) + R(0) = 1$$

3.2 Epidemic case

We first look at a case where an epidemic occurs. The parameters used are $\beta = 3, \gamma = 1$.

Using the equation from Section 2, the basic reproduction number is

$$R_0 = \frac{\beta}{\gamma} = \frac{3}{1} = 3.$$

Since $R_0 > 1$, we expect the number of infected people to increase at the start of the simulation.

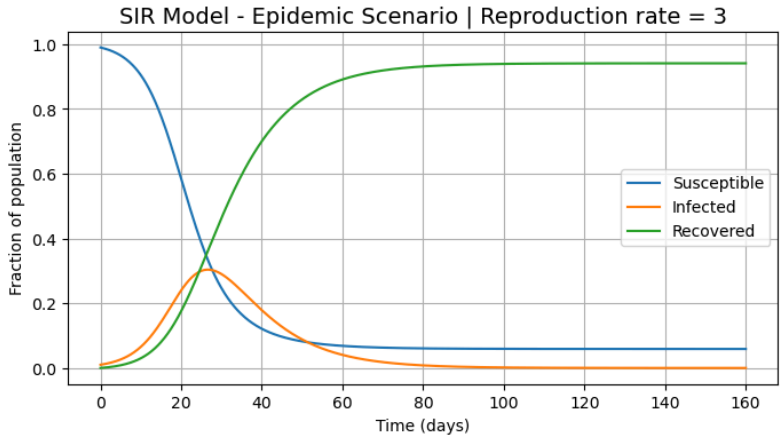


Figure 3. SIR model showing an epidemic with $R_0 = 3$.

As we expected, Figure 3 shows that the infected population starts to increase. At the beginning there are many susceptible people, so infected people have a high chance of passing on the disease. The number of infected people therefore rapidly increases.

As more people become infected, the susceptible population decreases and the recovered population increases. Eventually, the infected population reaches a peak and starts to decrease. At this point there are no longer enough susceptible people to pass the infection onto.

From Section 2, we know that the infected population reaches its peak when

$$S = \frac{\gamma}{\beta} = \frac{1}{R_0}$$

For this simulation,

$$S = \frac{1}{3} \approx 0.333$$

We therefore expect the peak of the infected population to occur when around 33.3% of the population is still susceptible. After this point, the recovery of infected people happens faster than the creation of new infections. Another important observation is that S does not reach zero. Some people remain susceptible even after the epidemic has ended.

3.3 Non-epidemic case

We can now compare the previous result with a case where an epidemic does not occur. We keep the recovery rate and initial conditions the same, but reduce the transmission rate:

$$\beta = 0.9, \gamma = 1.$$

The reproduction number is now

$$R_0 = \frac{0.9}{1} = 0.9.$$

Since $R_0 < 1$, we expect the disease to die out.

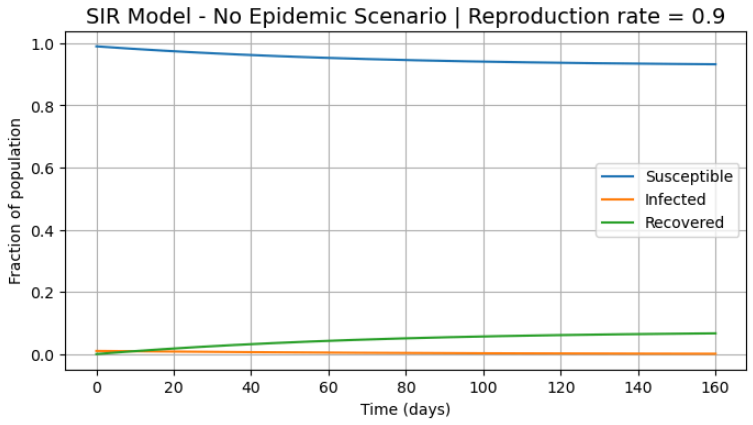


Figure 4. SIR model showing a non-epidemic with $R_0 = 0.9$.

Figure 4 shows a large difference compared with Figure 3. The infected population does not first increase to an epidemic peak. Instead, it decreases from the beginning of the simulation. This means that most of the population therefore remains susceptible and only a small fraction ever becomes infected.

The main difference between Figures 3 and 4 is the value of R_0. We only changed β, while keeping γ and the initial conditions are the same. This makes it easier to see how the transmission rate changes the behaviour of the model.

3.4 Phase-space analysis

We can also compare the two simulations using phase-space diagrams. Instead of showing the compartments as a function of time, we can plot the infected population I against the susceptible population S.

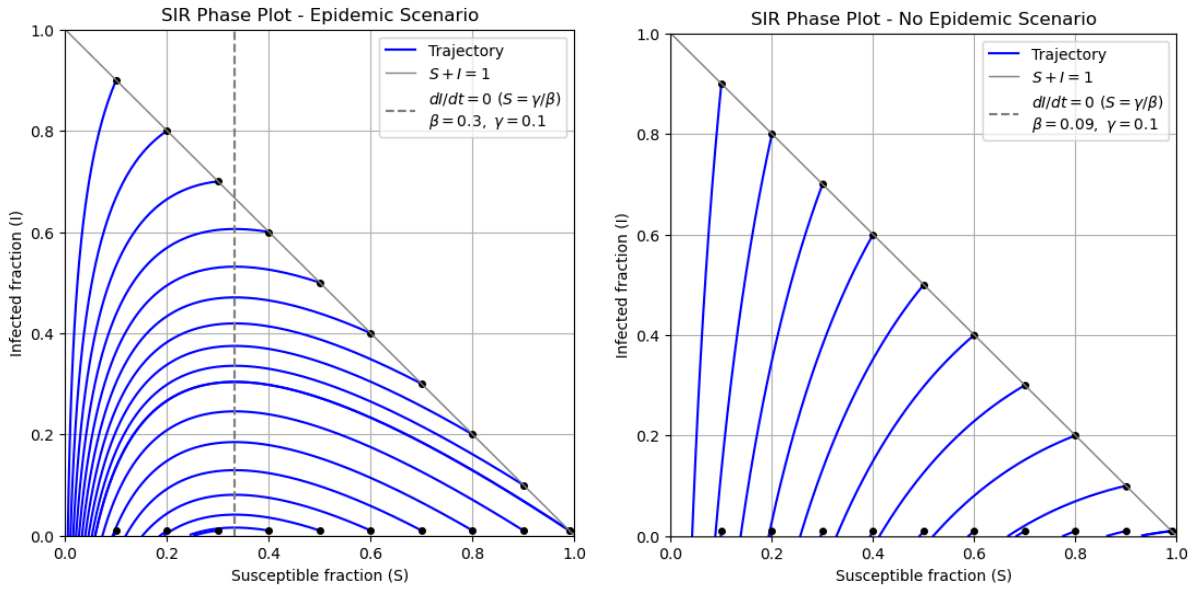


Figure 5. Phase-space plots of I against S for $R_0 = 3$ and $R_0 = 0.9$.

The starting conditions on the diagonal line $S + I = 1$ simulates different starting fraction of infected people. Starting points on the x-axis show a fraction of the population starting in the recovered class, this could be due to vaccination for example.

For $R_0 = 3$, the infected fraction first increases as the susceptible fraction decreases. The curve reaches its highest value of I around $S = 1/3$, which agrees with the condition derived earlier. After this point, both S and I decrease until the infected fraction approaches zero.

For $R_0 = 0.9$, there is no epidemic peak. The infected population starts decreasing immediately. The phase-space diagram therefore shows a much smaller change in S. This again shows that the disease is unable to spread through the population when $R_0 < 1$.

4 Fitting the SIR Model to Influenza Data

4.1 Influenza outbreak data

So far, we have tested the SIR model using parameters that we selected ourselves. We can now test how well the model can describe a real epidemic. For this, we use data from an influenza outbreak at a boys' school. The school had a total population of 763 boys and the outbreak started with one infected boy.

The number of infected boys was recorded each day for 14 days. The data is shown in Table 1.

Day	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Number	1	3	8	28	75	221	291	255	235	190	125	70	28	12	5

Table 1. Number of infected boys in the influenza outbreak

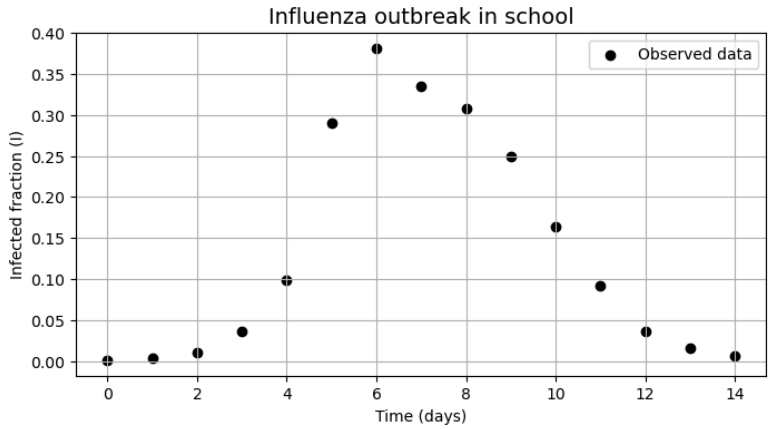


Figure 6. Influenza outbreak data plotted

From the data, we can already see the general shape of an epidemic. The number of infected boys first increases quickly, reaching a maximum of 291 on day 6. After this point, the number starts to decrease again. This is similar to the infected curve produced by the SIR model in Figure 3.

4.2 Finding the parameters

To fit the SIR model to the data, we need to find suitable values for the transmission rate β and recovery rate γ. We want to find the values that make the model as close as possible to the observed data.

The initial conditions of the model can be found directly from the school data. There are 763 boys and only one is infected at the beginning. We therefore use:

$$S(0) = \frac{762}{763} \approx 0.9987$$

$$I(0) = \frac{1}{763} \approx 0.0013$$

$$R(0) = 0$$

To find the optimal values for our parameters, we will simulate the infected curve many times and find the values that minimize the difference between the simulated SIR model and the observed data. To measure this difference, we use the sum of squared errors (SSE):

$$SSE = \sum_{t=0}^{14} \left(I_{simulated}(t) - I_{observed}(t) \right)^2$$

For every pair of β and γ, the SIR model produces a value for the infected fraction each day. We compare these values with the observed infected fractions. The difference at each day is squared and all of these values are then added together. Squaring the difference makes all errors positive and gives larger errors more importance. The optimisation gives the following parameters:

$$\beta \approx 1.67$$

$$\gamma \approx 0.45$$

$$R_0 = \frac{1.67}{0.45} \approx 3.71$$

Because $R_0 > 1$, we expect an outbreak to take place, which matches the recorded data from the school outbreak. We can also observe a relatively high infection rate β of 1.67, however the recovery rate is also high, taking just over 2 days to recover from this influenza.

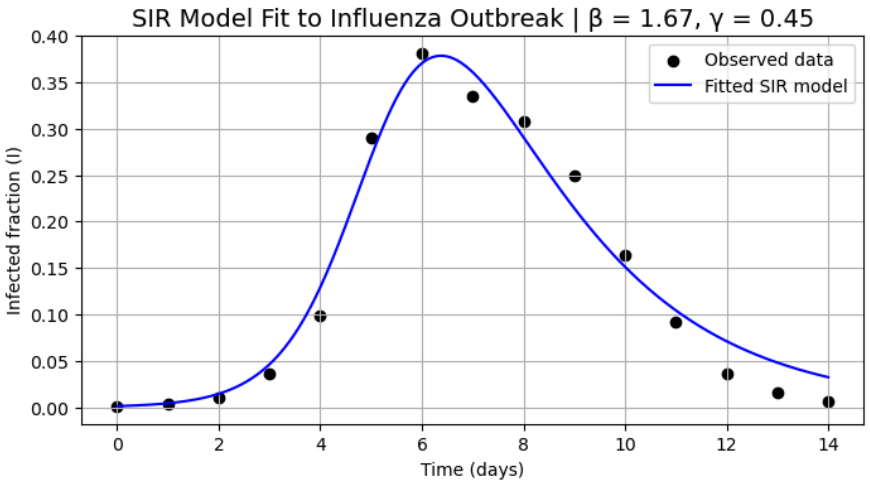


Figure 7. Fitted simulated curve onto the observed data

We can now compare the fitted model with the observed data. From Figure 7, we can see that the SIR model reproduces the general behaviour of the influenza outbreak quite well. The simulated number of infections increases rapidly during the first few days and reaches its peak at approximately the same time as the observed data. The model also reproduces the decrease in infections after the peak.

It is clear that the observed data do not follow a smooth curve that could be perfectly fit with fitted parameters. In real world situations, there are many more variables and factors that are not captured by this particular model. For example, the model assumes that all boys have the same probability of interacting with each other and that the transmission and recovery rates remain constant throughout the outbreak. Furthermore, tracking the number of infected people may not always be accurate, due to things such as noise and delayed symptoms.

Finding the most accurate parameters to the outbreak is crucial because it enables us to evaluate how certain preventions and strategies are effective at decreasing the outbreak. Two vaccination strategies were designed and assessed at stopping the epidemic.

5 Vaccination Strategies

We compare two strategies. In the first strategy, 10 susceptible students are vaccinated each day. This represents a gradual vaccination programme after the outbreak begins. In the second strategy, we use critical vaccination, where just enough of the population is vaccinated before the outbreak to prevent the disease from spreading.

Our hypothesis is therefore that vaccinating 10 students per day will reduce the maximum number of

infected boys and reduce the total number of infections. However, because vaccination happens only after the outbreak starts, we still expect an epidemic to occur before enough students have been vaccinated.

The second strategy is based on the reproduction number. An epidemic can initially spread when

$$R_0 S_0 > 1.$$

If a fraction p of the population is successfully vaccinated before the outbreak, the susceptible fraction becomes approximately

$$S_0 = 1 - p.$$

To prevent the epidemic from growing, we therefore require

$$R_0(1 - p) < 1.$$

Solving this for p gives the critical vaccination fraction

$$p > 1 - \frac{1}{R_0}.$$

For our fitted value of $R_0 = 3.71$,

$$p = 1 - \frac{1}{3.71} \approx 0.730.$$

Therefore, approximately 73.0% of the population would need to be vaccinated before the outbreak to bring the initial reproduction number below 1 under the assumptions of this model. For a school of 763 boys, this corresponds to 557 students.

Our hypothesis is that this strategy will prevent a epidemic from occurring because the susceptible population starts below the epidemic threshold.

We can now compare both of the vaccination strategies with the original SIR model.

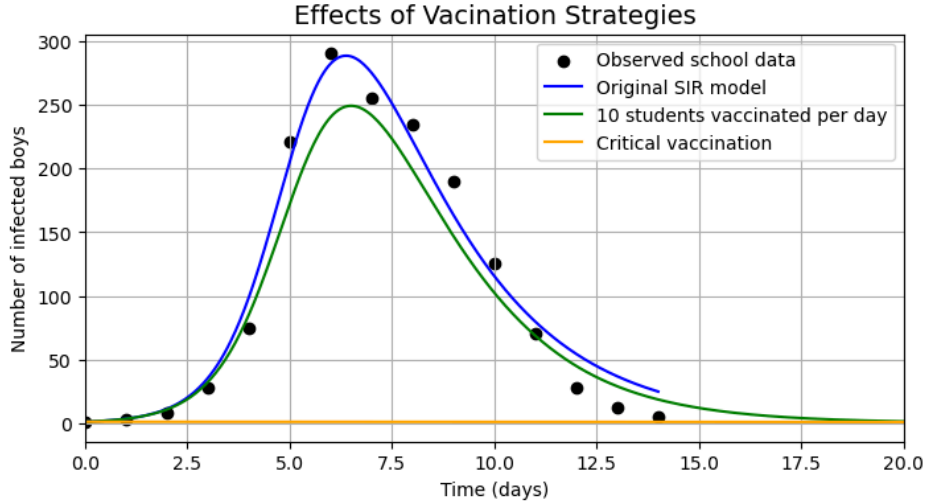


Figure 8. Comparing the effects of vaccination strategies.

Figure 8 shows a clear difference between the three cases. Without vaccination, the original fitted model reaches a peak of approximately 290 infected boys around day 6. This is close to the peak seen in the observed school data.

When 10 students are vaccinated per day, the peak is reduced to approximately 250 infected boys. The peak also occurs slightly later. This shows that even a gradual vaccination programme can reduce the size of the epidemic. However, a large outbreak still occurs because the disease spreads quickly during the first few days.

The critical vaccination strategy has a much stronger effect. In Figure 8, the infected population remains close to zero instead of developing into a large epidemic. This happens because enough students are already protected at the beginning of the simulation. The susceptible fraction is therefore below the threshold required for the infection to grow.

6 SIR Model with Demography

6.1 Adding births and deaths

The basic SIR model assumes that the total population is closed. However, over longer periods of time, births and natural deaths become important. We can therefore extend the model by introducing a demographic rate μ. We assume that the birth and natural death rates are equal, meaning that the total population remains constant.

The equations for the SIR model with demography are

$$\frac{dS}{dt} = \mu - \beta SI - \mu S$$

$$\frac{dI}{dt} = \beta SI - \gamma I - \mu I$$

$$\frac{dR}{dt} = \gamma I - \mu R$$

The term μ in the susceptible equation represents new individuals entering the population through births. These individuals are assumed to be susceptible. At the same time, individuals leave each of the three compartments through natural death at a rate proportional to the size of that compartment.

Since the birth and death rates are equal, the total population remains constant:

$$\frac{dN}{dt} = \frac{dS}{dt} + \frac{dI}{dt} + \frac{dR}{dt} = 0$$

Therefore,

$$S + I + R = 1$$

The important difference compared with the basic SIR model is that the susceptible population can now increase again. In the basic model, S can only decrease. With demography, births continuously introduce new susceptible individuals. This makes it possible for the disease to remain in the population instead of completely dying out.

6.2 Endemic equilibrium

Endemic equilibrium occurs when an infection persists within a population indefinitely at a non-zero level. To investigate this behaviour, we simulate the model over a longer period of 500 days.

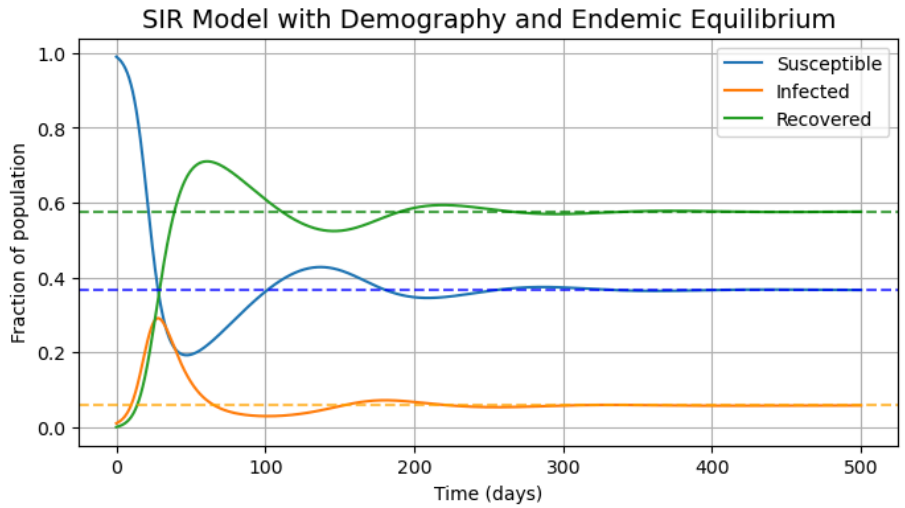


Figure 9. SIR model with demography and endemic equilibrium

Figure 9 shows behaviour that we did not observe in the basic SIR model. At the beginning, the number of infected individuals increases quickly and produces a large epidemic peak. The infected fraction then decreases, but instead of approaching zero, it begins to increase again. It proceeds to oscillate slightly towards a fixed point.

Eventually, all three compartments approach constant values. We can calculate each fraction of the population at the endemic equilibrium as following:

$$S^* = \frac{\gamma+\mu}{\beta} \approx 0.367$$

The infected fraction can then be calculated using

$$I^* = \frac{\mu(1-S^*)}{\gamma+\mu} \approx 0.058$$

and since the total population is equal to 1, the recovered fraction is

$$R^* = 1 - S^* - I^* \approx 0.576$$

Therefore, at the endemic equilibrium approximately 36.7% of the population is susceptible, 5.8% is infected and 57.6% is recovered. These three values are shown by the dashed horizontal lines in

Figure 9.

The main reason this is possible is the introduction of births. New susceptible individuals continuously enter the population. This creates new opportunities for the disease to spread and prevents the infected population from permanently disappearing. At equilibrium, the movement of individuals between the compartments balances out. Meaning that the fractions of susceptible, infected and recovered individuals remain constant.

6.3 Effect of the demography rate

We can now investigate how the demographic rate μ changes these oscillations. We run the model using four different values:

$$\mu = 0.040, 0.020, 0.013, 0.010.$$

The other parameters and initial conditions are kept the same.

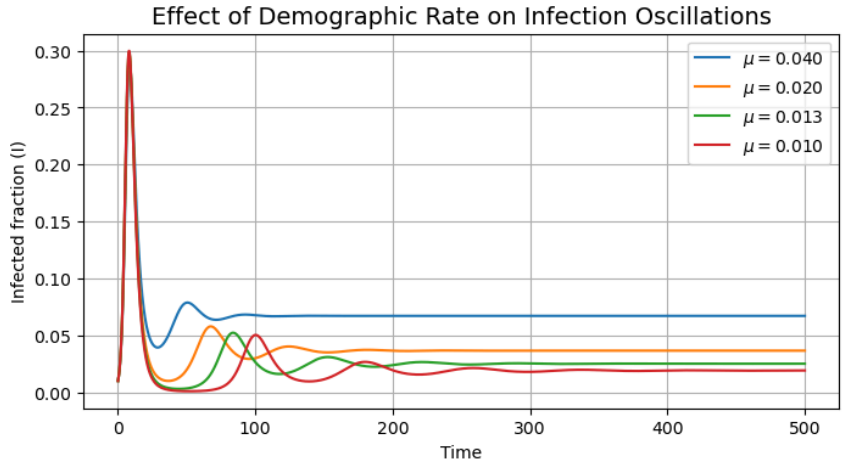


Figure 10. Infected fractions for different demographic rates μ

The figure above shows that changing μ affects both the oscillations and the final infected fraction. All four simulations have a similar first epidemic peak. After this peak, however, their behaviour becomes different.

For larger values of μ, susceptible individuals are replaced more quickly. The infected population therefore returns from its first minimum sooner. The oscillations also settle towards a higher endemic infected fraction. For example, $\mu = 0.040$ approaches an infected fraction of approximately 0.07, while $\mu = 0.010$ approaches a value closer to 0.02.

For smaller values of μ, the later infection peaks occur further apart. This is because susceptible individuals are added to the population more slowly. It therefore takes longer for a sufficiently large susceptible population to build up and allow the infected population to increase again.

An important observation is that these oscillations are damped. Their amplitude becomes smaller with time until the model approaches the endemic equilibrium. They are therefore different from permanent repeated epidemics which we will see later on in this report.

6.4 Phase-space behaviour

The behaviour of the model and its oscillations can be studied using a phase-space diagram. Figure 11 plots the infected fraction I against the susceptible fraction S for several different initial conditions.

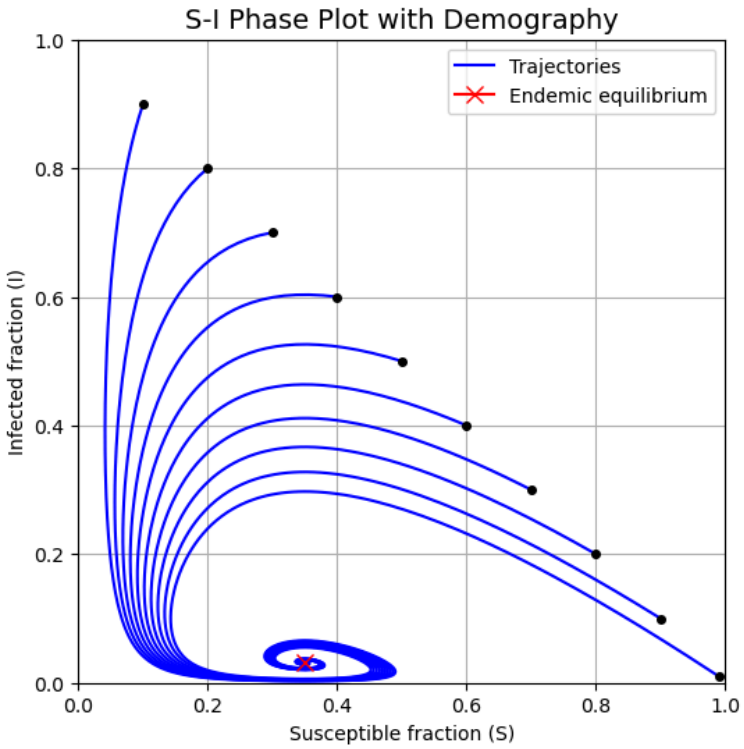


Figure 11. Phase-space diagram of the SIR model with demography.

Close to the endemic equilibrium, the trajectories form a spiral. Each loop around the equilibrium represents an oscillation in the susceptible and infected populations. As time increases, the loops become smaller and the trajectory approaches the equilibrium.

This also shows that the endemic equilibrium is stable for these parameter values. Even when the model begins with different susceptible and infected fractions, the trajectories approach the same equilibrium. The starting conditions mainly change the path taken towards the equilibrium rather than its final position.

6.5 Fourier analysis of the oscillations

The previous figures show that the infected population oscillates before reaching its equilibrium. We can use a Fourier transform to study the frequency of these oscillations.

A time series such as $I(t)$ shows how the infected population changes over time. A Fourier transform instead represents this signal in terms of its frequencies. A large peak in the Fourier spectrum means that a particular frequency is strongly present in the original signal.

Before calculating the Fourier transform, we subtract the mean infected fraction,

$$I'(t) = I(t) - \bar{I}$$

so that the constant endemic level does not dominate the zero-frequency component. We then calculate the discrete Fourier transform using the Fast Fourier Transform (FFT).

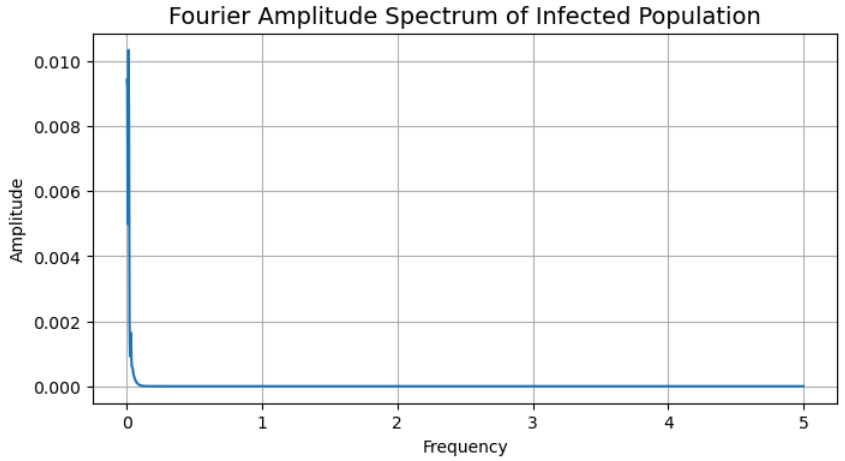


Figure 12. Fourier amplitude spectrum of the infected population in the SIR model with demography.

Figure 12 shows that all of the amplitude is concentrated at very low frequency. This agrees with the time-domain results because the oscillations take place over relatively long periods compared with the sampling interval.

The dominant non-zero frequency has a value of:

$$f_{dominant} = 0.01199$$

The corresponding period is:

$$T = \frac{1}{f_{dominant}} = 83.350\ days$$

This value can then be compared with the distance between the peaks in Figure 9. With the same initial conditions and parameters, we should expect a damping peak of the infected fraction of population every 83.35 days.

6.6 Infection-induced mortality

The previous model includes natural births and deaths while keeping the total population constant. However, an infectious disease can also cause additional deaths. We can therefore extend the model by introducing infection-induced mortality.

We use α to represent the infection-induced mortality rate. Meaning that a fraction of infected people will die instead of recover. $N = S + I + R$ represents the living population. Unlike the previous model, N is no longer constant because individuals who die from the infection leave the living population.

$$\frac{dN}{dt} = -\alpha I$$

Therefore, whenever both $I > 0$ and $\alpha > 0$, the total living population decreases.

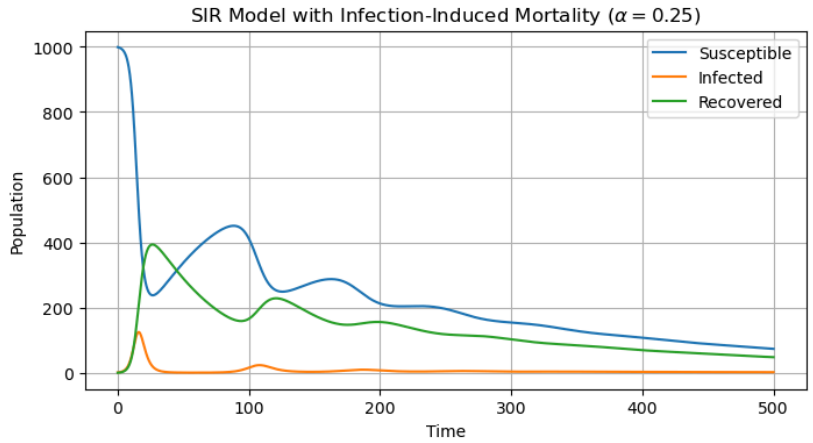


Figure 13. SIR model with Mortality ($\alpha = 0.25, N = 1000$)

Above is an example simulation of an epidemic that has a mortality rate of 0.25. We can observe an initial peak of infections and then what may appear as oscillations. An important difference from figure 9 is that the population fractions no longer approach the same constant endemic equilibrium. Instead, the sizes of the compartments decrease over time, dropping from a population of 1000 to around 200 after 500 days. We can now investigate what happens as we change α.

We compare: α = 0, 0.25, 0.5, 0.6, 0.8.

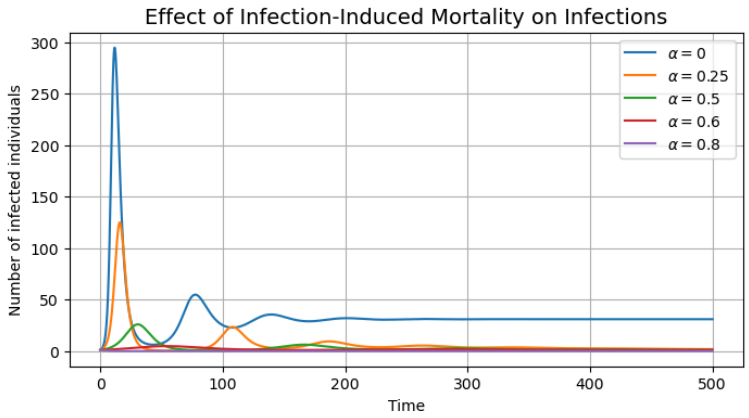


Figure 14. Number of infected people for different values of mortality rate α.

We can see that increasing α strongly changes the infection simulations. When $\alpha = 0$, there are no deaths caused by the infection. The first epidemic peak reaches almost 300 infected individuals and later approaches an endemic level of approximately 30 infected individuals. As you increase α , the first peak decreases drastically, to about 125 infected individuals at $\alpha = 0.25$, and to about 25 infected individuals at $\alpha = 0.5$.

It may appear as a higher mortality rate leads to a better outcome. However, the truth is that the infected population becomes smaller because infected individuals are more likely to die and leave the living population. Next we investigate how the living population is affected by mortality rate.

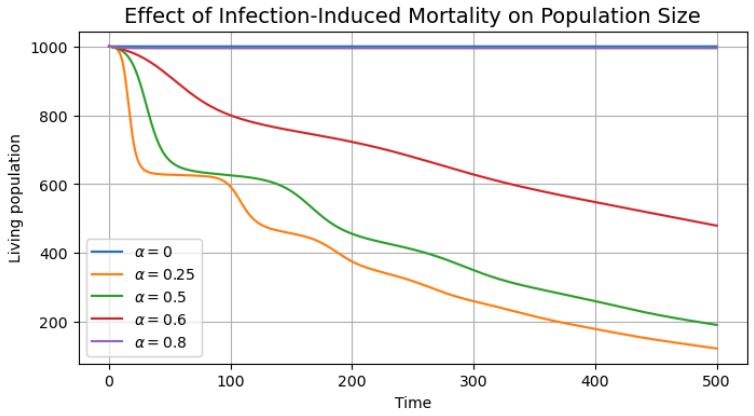


Figure 15. Total living population for different mortality rates.

Out of the 5 values tested for α, $\alpha = 0.25$ lead to the biggest decrease of the living population. Interestingly, the relationship between α and total population loss is not linear in our results. For example, the curve for $\alpha = 0.6$ remains above those for $\alpha = 0.25$ and $\alpha = 0.5$. A higher probability of death makes the infection more harmful to each infected individual, but it also changes the epidemic itself. When infected individuals are removed more quickly, there are fewer infected individuals available to continue transmission.

This becomes especially clear for $\alpha = 0.8$. In our simulation, the infection remains very small and the population stays close to its initial value. Therefore, as α approaches 1, we cannot simply say that a larger value always produces more total deaths. The mortality parameter also changes how successfully the infection can spread through the population.

7 SIR model variant

7.1 SEIR model

The SIR model assumes that a susceptible individual becomes infectious immediately after becoming infected. For many diseases this is not the case, since there can be a period between becoming infected and becoming infectious. We can account for this by adding another compartment to the model.

The SEIR model divides the population into susceptible (S), exposed (E), infected (I), and recovered (R). When a susceptible individual becomes infected, they first enter the exposed compartment. During this period, they have been infected but are not yet infectious. After the exposed period, they move into the infected compartment and can transmit the disease.

We use σ as the rate at which exposed individuals become infectious. Therefore, $\frac{1}{\sigma}$ represents the average amount of time an individual spends in the exposed compartment.

The ordinary differential equations for this model are as following:

$$\frac{dS}{dt} = \mu - \beta SI - \mu S$$

$$\frac{dE}{dt} = \beta SI - \sigma E - \mu E$$

$$\frac{dI}{dt} = \sigma E - \gamma I - \mu I$$

$$\frac{dR}{dt} = \gamma I - \mu R$$

As before, births enter the susceptible compartment and natural deaths occur in every compartment. The total population therefore remains constant.

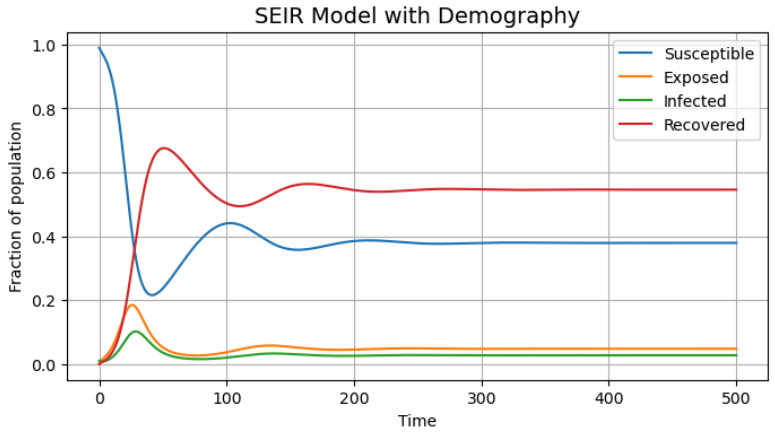


Figure 16. Example of an SEIR model simulation with demography.

Figure 16 shows similar general behaviour to the SIR model with demography, but there is now an additional exposed population. The following parameter values have been used:

$$\beta = 1, \gamma = \frac{1}{3}, \mu = \frac{1}{60}, \sigma = \frac{1}{5}$$

And these initial conditions have been chosen:

$$\mathbf{y}_0 = \begin{bmatrix} S(0) \\ E(0) \\ I(0) \\ R(0) \end{bmatrix} = \begin{bmatrix} 0.99 \\ 0 \\ 0.01 \\ 0 \end{bmatrix}$$

7.2 Effect of the exposed period

The new parameter σ allows us to investigate how the length of the exposed period changes the epidemic. We compare four average exposed periods:

$$\frac{1}{\sigma} = 2, 5, 10, 20$$

All other parameters and initial conditions are kept the same.

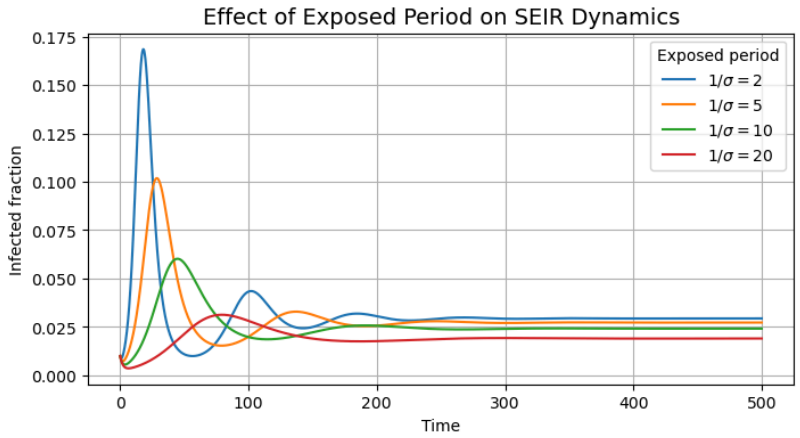


Figure 17. Effect of different exposed periods on the infected fraction.

Figure 17 shows that the exposed period has a large effect on both the timing and size of the first epidemic peak. When $\frac{1}{\sigma} = 2$, individuals remain exposed for only a short period. The infected population therefore increases quickly and reaches a large peak of approximately 0.170.

As the exposed period increases, the epidemic becomes slower. For $1/\sigma = 5$, the peak occurs later and is reduced to approximately 0.10. At $1/\sigma = 10$, the peak moves even further to the right and falls to approximately 0.06.

We can therefore see that increasing the exposed period delays and reduces the epidemic peak. Individuals remain in the exposed compartment for longer before they can transmit the disease.

We can next analyse the endemic states the model approaches from different initial conditions. We can again use a phase-diagram to investigate this behaviour.

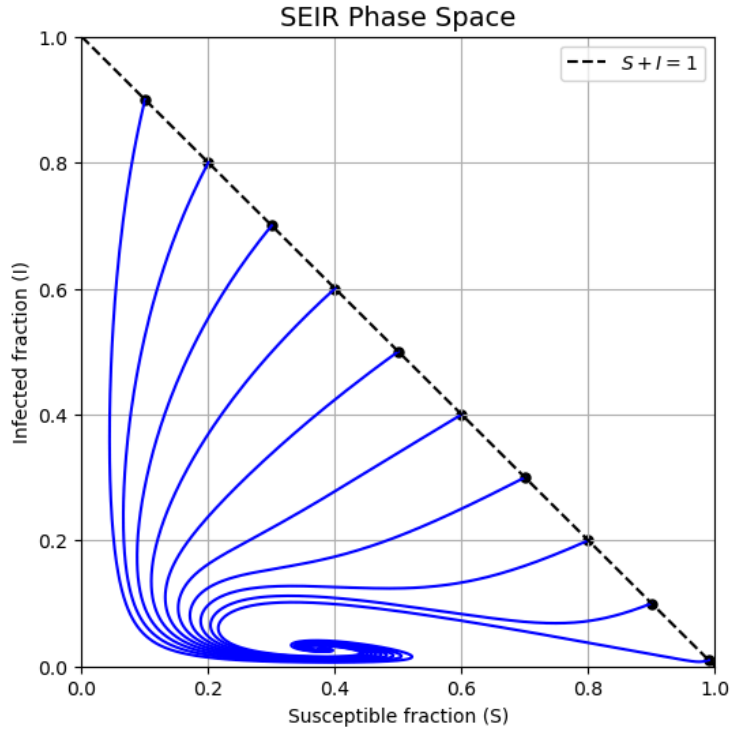


Figure 18. S - I Phase-space diagram.

Multiple trajectories are displayed starting from different initial conditions that satisfy S + I = 1. Even though the trajectories begin at very different points, they eventually move towards the same small region of the phase space.

The phase-space diagram therefore shows that the long-term behaviour is not strongly dependent on the initial values used here. Different starting conditions change the initial outbreak, but the trajectories eventually approach the same endemic equilibrium.

7.3 Seasonal Forcing of the SEIR Model

So far, we have assumed that the transmission rate β remains constant over time. In reality, contact between people can change throughout the year. For example, seasons, the start of school years, or holidays may increase interaction between certain groups of people.

We can represent this by making the transmission rate dependent on time. We use the following function:

$$\beta(t) = \beta_0\left[1 + \beta_1 \cos\left(\frac{2\pi t}{T}\right)\right]$$

where β_0 is the average transmission rate, T is the period of the seasonal cycle, and β_1 controls the strength of the seasonal forcing. We first compare four different strengths of seasonal forcing:

$$\beta_1 = 0, 0.2, 0.5, 0.8$$

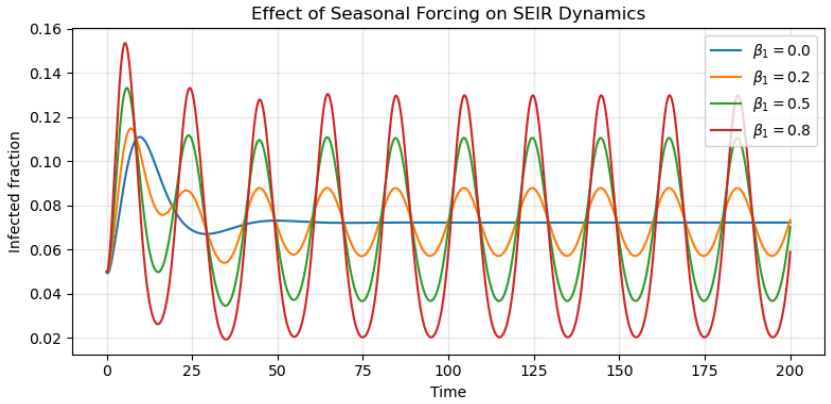


Figure 19.Effect of β_1 on the infected fraction of the SEIR model.

The main difference between this figure and the previous ones is that when seasonal forcing is introduced, the oscillations continue instead of disappear [2]. For $\beta_1 = 0.2$, the infected fraction shows relatively small repeated peaks around its endemic level. Increasing the forcing to $\beta_1 = 0.5$ produces larger oscillations. Changing the strength of the seasonal forcing does not impact the length or frequency of the outbreaks, only the fraction of infected individuals.

We can also observe how other compartments oscillate over longer periods of time.

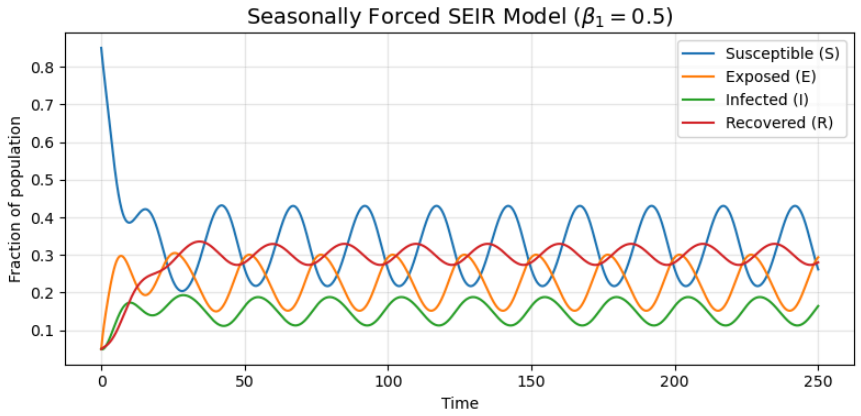


Figure 20. Seasonally forced SEIR model simulation

After the initial outbreak, the model settles into a repeating pattern rather than a constant equilibrium. The susceptible population increases and decreases as the transmission rate changes. When transmission increases, more susceptible individuals move into the exposed compartment. The exposed population therefore increases first, followed by the infected population. The recovered population changes more slowly because individuals must pass through both the exposed and infected compartments before recovering.

Comparing Figure 20 with Figure 16 shows the main effect of seasonal forcing. Without seasonal forcing, the oscillations become smaller until the system reaches an endemic equilibrium. With seasonal forcing, repeated changes in the transmission rate maintain the oscillations.

Overall, the SEIR model shows how adding one extra stage of infection can change the timing of an epidemic. The exposed period delays the growth of the infected population and can reduce its initial peak. Adding seasonal transmission changes the long-term behaviour by producing repeated epidemic waves. This shows how adding more variants to the basic SIR model can create much more complex disease dynamics.

8 Conclusions

In this report, we investigated how the SIR model can be used to simulate the spread of an infectious disease and how its behaviour changes when more complexity is added. The basic SIR model showed that R_0 is an important value for determining whether an epidemic can occur. We also fitted the model to influenza data from a boys' school and found $\beta = 1.67, \gamma = 0.45$, giving $R_0 \approx 3.71$. The vaccination experiments then showed how reducing the susceptible population can lower the size of an outbreak or prevent it from occurring.

Adding demography changed the long-term behaviour of the model. Births continuously introduced new susceptible individuals, which allowed the disease to reach an endemic equilibrium and created damped oscillations.

Finally, we extended the model to SEIR by adding an exposed compartment. A longer exposed period delayed the epidemic and reduced its initial peak. Adding seasonal forcing produced repeated epidemic waves instead of the model settling at a constant equilibrium.

Our results show that even the basic SIR model can reproduce important features of an epidemic. Adding processes such as demography, an exposed period and seasonal transmission can produce more realistic and complex behaviour. This also shows the importance of understanding the assumptions behind a model when using it in real world disease outbreaks.

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